motion INFORMATICS

FAQ: Frequently Asked Questions



FDA CLEARED

Contact us today: wecare@motioninformatics.ai | www.motioninformatics.ai

Frequently Asked Questions

- **What is Stimel-03?** Stimel-03 is a medical device integrating FES, NMES, and biofeedback technologies to provide comprehensive muscle re-education and pain management for patients in rehabilitation.
- **What is BioRhythmIQ Technology?** BioRhythmIQ is a patented technology integrated into the Stimel-03 device, designed to enhance rehabilitation therapies. It captures, cleans, and processes patient-specific neuromotor signals, ensuring precise synchronization with the patient's natural neuromuscular rhythms. This technology optimizes Functional Electrical Stimulation (FES), Neuromuscular Electrical Stimulation (NMES), and biofeedback therapies by personalizing and refining the electrical inputs based on individual patient needs. By doing so, BioRhythmIQ significantly improves the effectiveness of these therapies in restoring motor functions, enhancing muscle strength, and accelerating overall recovery, making it a crucial component in neuromuscular rehabilitation.
- **How does Stimel-03 work?** The device delivers electrical impulses to stimulate muscles and nerves, promoting muscle strength, coordination, and pain relief. Biofeedback enhances patient awareness and control over their physiological processes.
- **What are the primary applications of Stimel-03?** It is used in rehabilitation for patients recovering from strokes, injuries, or surgeries requiring muscle re-education, improved motor control, and pain management.
- **Who can benefit from using Stimel-03?** Patients with neurological and musculoskeletal conditions, including stroke survivors, individuals with spinal cord injuries, and those undergoing orthopedic rehabilitation.
- **How does Stimel-03 ensure compliance with FDA regulatory standards?** Stimel-03 is FDA cleared under 510(k) (K 130424), adhering to specific regulatory standards that confirm its safety and efficacy for intended medical uses, including neuromuscular stimulation.
- **What makes Stimel-03 a preferred device among physical therapists?** Physical therapists prefer Stimel-03 due to its effectiveness in enhancing muscle function, ease of use, and the ability to provide targeted, patient-specific treatments that improve rehabilitation outcomes.
- **How does Stimel-03 support therapists in private clinics?** It provides therapists in settings like physiotherapy, sports rehabilitation, and orthopedics with advanced tools like NMES and FES, enabling effective patient management and personalized treatment protocols.
- **How does the compact design of Stimel-03 benefit its usage in multiple settings?** Its compact design makes it easy to transport and set up in various clinical, home, and field settings, allowing for versatile use without compromising on functionality or effectiveness.
- **How durable is the Stimel-03 device?** Designed for both clinical and intensive home use, the device is built to withstand regular use under various conditions.
- **Is Stimel-03 suitable for home use?** Yes, it is designed for both clinical and home settings, offering flexibility in rehabilitation programs.
- **What makes Stimel-03 suitable for use by patients at home?** Stimel-03's user-friendly interface and safety features allow patients to use the device at home with minimal supervision, supporting continued care and facilitating regular therapy sessions as prescribed.
- **What makes Stimel-03 different from other similar devices?** Its unique integration of FES, NMES, and biofeedback in one device offers a more holistic approach to rehabilitation not commonly found in other devices.

- **Can Stimel-03 be customized for individual patient needs?** Yes, it offers highly customizable settings for intensity, frequency, and duration of stimulation to suit individual therapy needs.
- **How does Stimel-03 enhance patient compliance?** The device's user-friendly interface and real-time feedback help patients understand their progress and engage more actively in their rehabilitation.
- **What safety features does Stimel-03 include?** It includes automatic shut-off, adjustable intensity limits, and real-time monitoring to ensure patient safety during use.
- **What are the key differentiators of Stimel-03 from other devices in the market?** Key differentiators include its integrated use of NMES, FES, and biofeedback, personalized medicine approach with BioRhythmIQ technology, and its comprehensive application across various severities of neuromuscular conditions.
- **How does Stimel-03's approach to biofeedback differ from other biofeedback devices?** Stimel-03's biofeedback mechanism is integrated with NMES and FES, offering real-time feedback that is more closely aligned with the patient's physiological responses and rehabilitation needs.
- **How does biofeedback help in rehabilitation with Stimel-03?** Biofeedback provides patients with real-time data about their muscle activity, enabling them to make adjustments to improve their muscle control and effectiveness of the therapy.
- **What is the recommended duration and frequency of therapy sessions with Stimel-03?** Typically, therapy sessions last between 20 to 30 minutes and are recommended 2-3 times a week, depending on the patient's individual rehabilitation plan.
- **Can Stimel-03 be used for pediatric patients?** Yes, it can be used for pediatric patients, but settings should be carefully adjusted under the supervision of a qualified therapist.
- **Is there training available for clinicians new to Stimel-03?** Yes, comprehensive training for clinicians is provided to ensure they are well-equipped to use the device effectively with patients.
- **Are there any contraindications or precautions associated with using the Stimel-03?** Contraindications for using the Stimel-03 include patients with pacemakers or other electronic implanted devices, as well as those with certain types of cancer. Precautions must be taken for patients with cognitive impairments or severe skin conditions. It is important for clinicians to evaluate each patient's specific health conditions and medical history before initiating treatment with the Stimel-03.
- **What improvements in function can patients expect from using Stimel-03?** Patients can expect improvements in muscle strength, coordination, pain levels, and overall mobility.
- **Can Stimel-03 be integrated with other therapeutic modalities?** Yes, it can be used in conjunction with other therapeutic approaches like manual therapy and exercise to enhance overall treatment efficacy.
- **Is there a warranty or service plan available?** The device comes with a standard 12-month warranty, and extended service plans are available to cover maintenance and repairs.
- **How does Stimel-03 improve clinical efficiency?** Its comprehensive capabilities allow clinicians to treat multiple aspects of patient care with one device, improving treatment efficiency and outcomes.
- **What feedback mechanisms are included to monitor patient progress?** The device provides detailed feedback on muscle activation patterns and progress metrics through its software interface.
- **Can Stimel-03 be used immediately post-surgery?** Use post-surgery depends on individual cases and should be approved by the treating physician to ensure safety.
- **How cost-effective is using Stimel-03 in a rehabilitation setting?** Its versatility and efficacy in treating various conditions make it a cost-effective solution, reducing the need for multiple devices.

- **How cost-effective is the Stimel-03 in comparison to traditional rehabilitation methods?** The Stimel-03 is cost-effective compared to traditional rehabilitation methods as it can reduce the overall duration of therapy needed by enhancing the efficiency of each session. It supports faster muscle recovery and motor control, potentially reducing the number of therapy sessions required and allowing for more patients to be treated in a shorter time frame.
- **How does Stimel-03 contribute to reducing healthcare costs?** By enhancing the effectiveness of rehabilitation sessions and reducing the time required for recovery, Stimel-03 can potentially lower overall healthcare costs associated with long-term rehabilitation.
- **What are the storage and care instructions for the Stimel-03?** The device should be stored in a dry, cool place and cleaned according to the instructions provided to ensure longevity and hygiene.
- **What are the infection control protocols for using Stimel-03 in a clinical setting?** Infection control protocols include regular cleaning and disinfection of the device and accessories, using disposable or sterilizable electrodes, and adhering to hygiene standards.
- **What clinical evidence supports the efficacy of Stimel-03 in stroke rehabilitation?** Clinical studies have shown that the integration of FES, NMES, and biofeedback can significantly improve muscle strength and function in stroke patients, leading to enhanced mobility and independence.
- **How does Stimel-03 modulate its electrical stimulation to prevent muscle fatigue during therapy sessions?** The device uses varying frequencies and intensities to ensure muscles are stimulated without causing excessive fatigue, optimizing recovery and endurance over time.
- **What are the specific parameters of NMES provided by Stimel-03 for muscle re-education?** NMES settings on Stimel-03 include adjustable parameters like pulse width, frequency, and intensity to target specific muscle groups effectively for re-education purposes.
- **Can Stimel-03 be used to treat spasticity in patients with neurological disorders?** Yes, the device can help manage spasticity by alternating muscle contractions and relaxations, improving range of motion and reducing stiffness.
- **How does the biofeedback feature enhance the treatment outcomes in patients using Stimel-03?** Biofeedback provides real-time visual feedback on muscle activity, helping patients and therapists optimize the use of targeted muscles during therapy sessions.
- **What are the best practices for integrating Stimel-03 into a post-operative recovery protocol?** Best practices include starting with mild stimulation settings, closely monitoring patient comfort, and gradually increasing intensity as recovery progresses.
- **How does Stimel-03 handle patients with varying levels of sensitivity to electrical stimulation?** The device offers extensive customization of stimulation parameters, allowing therapists to fine-tune the therapy to match individual patient sensitivity and tolerance.
- **Can Stimel-03 be used in conjunction with pharmacological treatments for pain management?** Yes, it can be used as an adjunct therapy to pharmacological treatments, providing a synergistic effect that may reduce the need for high dosages of pain medications.
- **What impact does Stimel-03 have on joint mobility and flexibility in arthritic patients?** Regular use of NMES and FES can help maintain and improve joint mobility and flexibility by stimulating muscle movement and reducing pain around the joints.
- **What guidelines should be followed when using Stimel-03 on patients with dermatological conditions?** For patients with skin conditions, it's crucial to use hypoallergenic electrodes, apply them to unbroken skin, and frequently inspect the skin for any signs of irritation or adverse reactions.

- C**an Stimel-03 be used immediately post-surgery? What are the benefits?** Yes, it can be used for immediate post-surgical stimulation of calf muscles to prevent venous thrombosis, promoting muscle contractions and enhancing blood flow to prevent clot formation.
- **What type of battery powers Stimel-03, and what are its benefits?** Stimel-03 is powered by a rechargeable Nickel Metal Hydride (Ni-MH) battery, providing reliable and long-lasting power for continuous and effective therapy sessions.
- **How does the LED bar display on Stimel-03 enhance treatment?** The LED bar display provides visual feedback that helps patients and therapists monitor the intensity and effectiveness of therapy, ensuring that each session is conducted within optimal therapeutic parameters.
- **What research supports the design and functionality of Stimel-03?** The design and functionality of Stimel-03 are supported by extensive clinical research that shows improved recovery outcomes in neuromuscular rehabilitation, validated through FDA clearance and clinical trials.
- **How does Stimel-03 facilitate muscle contraction during therapy sessions?** It uses modulated bipolar stimulation currents, either simultaneous or alternating, to effectively provoke muscle contractions, aiding in muscle re-education and strength rehabilitation.
- **What advancements does Stimel-03 bring to traditional NMES and FES therapies?** Stimel-03 enhances traditional therapies by incorporating BioRhythmIQ technology, which personalizes electrical stimulation based on patient-specific neuromotor signals, improving the precision and effectiveness of treatments.
- **How does Stimel-03 handle patient-specific thresholds in stimulation?** It uses threshold units derived from the patient-specific thresholds for stimulation signal amplitude, providing personalized muscle contraction control that optimizes therapy efficacy for individual patients.
- **Can Stimel-03 be integrated with other therapeutic modalities for a holistic approach?** Yes, it can be integrated with other therapeutic modalities such as manual therapy and exercise, enhancing the overall rehabilitation program and offering a more holistic approach to patient recovery.
- **What clinical outcomes have been observed with the use of Stimel-03 in stroke rehabilitation?** Clinical outcomes include improved motor function, increased muscle strength and coordination, enhanced range of motion, and significant strides in achieving functional independence.
- **How does Stimel-03 ensure patient comfort during therapy?** The device features adjustable intensity levels and pulse parameters that can be tailored to each patient's comfort threshold, ensuring a pain-free and comfortable therapy experience.
- **What are the key research areas for future development of Stimel-03?** Future research areas include exploring additional therapeutic applications, enhancing the integration of biofeedback, and improving patient interface for easier self-management of therapy sessions. The Spatial StimelMD which will hit the market in 2026 is generated by AI, AR and will have digital and tele-medicine functions.
- **How does the Stimel-03 utilize mirror neuron activation to enhance rehabilitation outcomes?** The Stimel-03 leverages the activation of mirror neurons through its combination of Functional Electrical Stimulation (FES) and Neuromuscular Electrical Stimulation (NMES) alongside biofeedback. This approach helps simulate the neural activity involved in motor actions by engaging mirror neurons when patients observe actions performed either by themselves or by others, which is crucial for motor learning and rehabilitation, especially in stroke recovery.
- **What specific benefits does mirror neuron activation provide in the context of using Stimel-03?** Activating mirror neurons with the Stimel-03 can enhance both the cognitive and emotional aspects of rehabilitation. By observing movements and participating in

biofeedback-enhanced exercises, patients can improve their motor skills and emotional understanding, which aids in a more holistic recovery process. This is particularly beneficial in therapies involving complex social interactions and emotional responses, where understanding and empathy are developed.

- **Can the Stimel-03's mirror neuron stimulation improve recovery speeds for stroke patients?** Yes, the Stimel-03's use of mirror neuron stimulation, particularly through visual and proprioceptive feedback, has been shown to potentially increase the rate of motor recovery in stroke patients. This is because the activation of mirror neurons through observation and subsequent stimulation helps re-establish neural pathways more efficiently, leading to quicker recovery of motor functions.
- **What makes the Stimel-03's approach to integrating mirror neuron theory unique in rehabilitation technologies?** The Stimel-03 uniquely integrates Functional Electrical Stimulation, Neuromuscular Electrical Stimulation, and biofeedback to actively involve mirror neurons in the rehabilitation process. This not only aids in motor recovery but also enhances cognitive and emotional rehabilitation. The device's ability to mimic observed actions and provide real-time feedback helps align the body's natural recovery processes with neurological rehabilitation goals, making it a comprehensive tool for neuromuscular recovery.
- W**hat is biofeedback in the context of the Stimel-03 and how does it facilitate stroke rehabilitation?** Biofeedback in the Stimel-03 involves using electromyography (EMG) to monitor the electrical activity of muscles and motor neurons. This feedback loop is displayed visually on the device as patients attempt to activate their muscles. It enables patients to understand and respond to their own muscle activity, helping to retrain their brain's neural pathways and facilitating the recovery of motor functions post-stroke.
- How **does visual biofeedback contribute to neural relearning in patients using the Stimel-03?** The visual biofeedback system in Stimel-03 displays a mobile line on the device's screen, which moves towards a target line as the patient attempts to engage their muscles. This system captures faint neural signals from the brain, amplifies them, and feeds them back to the patient's muscles, allowing the patient to observe the corresponding limb movement. This process helps in neural relearning by enabling mirror neurons to learn and teach motor functions, reshaping the brain's learning centers through neuroplasticity.
- **Can you explain the role of mirror neurons in the biofeedback process of the Stimel-03?** Mirror neurons play a critical role in the biofeedback process of the Stimel-03 by mimicking visual movements observed by the patient. As patients watch their limbs move in response to their brain's commands, these neurons help facilitate the learning and memorization of new motor functions. This enhances the brain's ability to rewire itself and establish new neural pathways, compensating for those lost during a stroke or brain injury.
- **What makes the biofeedback mechanism of the Stimel-03 effective for patients with various neurological impairments besides stroke?** The biofeedback mechanism of the Stimel-03 is effective for various neurological impairments due to its capacity to capture, analyze, and enhance faint neural signals specific to each patient's unique movement signature. This individualized approach allows for tailored rehabilitation that can adjust to the specific recovery needs of patients with different conditions, such as car accident injuries or sports-related neurological impairments. The system's ability to filter out noise and focus on meaningful neural activity ensures that each session is optimized for maximum recovery potential.
- **What is biofeedback in the context of the Stimel-03 and how does it facilitate stroke rehabilitation?** Biofeedback in the Stimel-03 involves using electromyography (EMG) to monitor the electrical activity of muscles and motor neurons. This feedback loop is displayed visually on the device as patients attempt to activate their muscles. It enables

patients to understand and respond to their own muscle activity, helping to retrain their brain's neural pathways and facilitating the recovery of motor functions post-stroke.

- **What factors should be considered when selecting a specific Stimel-03 program for a patient recovering from a stroke?** When selecting a Stimel-03 program, it is crucial to consider the patient's specific neuromuscular needs, the stage of their recovery, and their sensitivity to neuromuscular stimuli. The device offers various programs tailored to different levels of paralysis, sensory awareness, and motor control. For example, patients with mild paralysis might start with Biofeedback 1, which offers gentle biofeedback for muscle control re-learning, while those with more significant recovery might benefit from higher thresholds like in Biofeedback 3. It's also important to assess the patient's pain tolerance and any hypersensitivity issues, as programs like the Sensitive program are designed for those with high sensitivity.
- **How does the progression from one Stimel-03 program to another support a patient's rehabilitation journey?** The progression between Stimel-03 programs is designed to build upon the patient's growing neuromuscular control and sensory awareness. For example, a patient may start with a program offering basic biofeedback and, as they gain muscle control and reduce spasticity, move to more advanced biofeedback settings. This staged approach ensures that each phase of the rehabilitation is tailored to the patient's evolving capabilities, gradually enhancing their motor skills and sensory perception. The guidance for progression is clearly defined for each program, ensuring that transitions are timely and align with the patient's recovery milestones.
- **How does the Stimel-03 integrate with other rehabilitation therapies?** The Stimel-03 is designed to complement traditional rehabilitation therapies by enhancing neuromuscular control and sensory perception through targeted electrical stimulation and biofeedback. It can be seamlessly integrated into a patient's broader rehabilitation program, supporting and reinforcing the gains made in physical, occupational, and speech therapy. This multidisciplinary approach can accelerate recovery and improve overall outcomes.
- **What are the criteria for determining when a patient should transition between different levels of biofeedback programs?** Transitioning between different biofeedback levels in the Stimel-03 is guided by specific clinical markers such as improvement in voluntary muscle control, reduction in spasticity, and enhanced sensory perception. Clinicians will typically move a patient to a more advanced program once they consistently achieve predefined objectives in muscle activation, strength, and control, documented during therapy sessions.
- **What patient feedback has been received regarding the use of the Stimel-03?** Patient feedback on the Stimel-03 has generally been positive, with many reporting faster improvements in muscle control and a high level of satisfaction with the personalized nature of the therapy. Patients appreciate the device's ability to provide real-time feedback, which helps them stay engaged and motivated throughout their rehabilitation process.
- **What specific settings on Stimel-03 can be adjusted by therapists?** Therapists can adjust settings such as pulse frequency (30-80 Hz), amplitude (0.1-75 mA), pulse series duration (1-10 seconds), and pulse-to-pause ratios (1:1 to 1:5) to tailor the therapy according to individual patient needs.
- **A physician used the Stimel 3 device on a female patient with arm and finger paralysis, along with finger spasticity, using the Biofeedback program. Initially, the treatment worked well, but after 2 minutes, the EMG level dropped to 0 and no stimulation was delivered. When initiating a new session, the same happened: stimulation worked well in the beginning and stopped after 2 minutes. Why did this happen?** One possible explanation is that the patient became fatigued and was no longer able to reach the EMG threshold level required to trigger stimulation. Recall that the EMG level increases gradually during treatment to motivate increased effort by the patient and achieve better performance. When the threshold value was lowered manually, stimulation resumed. This

suggests that adjusting the threshold can help maintain effective treatment in cases of fatigue or limited muscle activation.

- **What is the recommended duration for a treatment session?** The system allows for sessions ranging from 5 to 35 minutes. If the patient is capable, a 30-minute session is recommended for optimal results.